Abstract

This project presents the simulation and analysis of a single-phase SPWM inverter in MATLAB Simulink using an H-bridge configuration. The effects of modulation index, varied from 0.1 to 1.5, and switching frequency, tested at 5 kHz, 10 kHz, 15 kHz, 20 kHz, and 25 kHz, on the output voltage and Total Harmonic Distortion are examined. The results show that THD decreases as the modulation index approaches 1.0, beyond which overmodulation causes waveform clipping and increases distortion. Higher switching frequencies further improve output quality by shifting harmonics away from the fundamental 50 Hz component. The best performance was observed at a modulation index of 1.0 with a 25 kHz carrier, yielding a THD of 2.66% and a fundamental voltage of approximately 319 V. For practical applications, a switching frequency of 20 kHz provides a suitable compromise with a THD of 2.95%.

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Chapter 1

Introduction

Inverters are essential in power electronics, converting DC into AC for applications such as renewable energy systems, UPS, and motor drives. Among them, single-phase H-bridge inverters are widely used due to their simple structure and effective control of output voltage.

In this work, Sinusoidal Pulse Width Modulation (SPWM) is implemented in MATLAB/Simulink to generate a near-sinusoidal AC output. The inverter output is controlled by the modulation index M_a, defined as:

$$M_a = \frac{V_{\text{ref}}}{V_{\text{carrier}}} \tag{1.1}$$

By varying M_a and the PWM switching frequency, the output voltage quality and Total Harmonic Distortion (THD) are analyzed to determine optimal operating conditions.

1.1 Generation of Pulse Width Modulation (PWM)

A PWM signal is created by comparing a low-frequency reference signal with a high-frequency carrier wave, like a sawtooth. When the reference signal is lower than the carrier, the output is 'High'; when it is higher, the output is 'Low.' The width of each pulse depends on the amplitude of the reference signal.

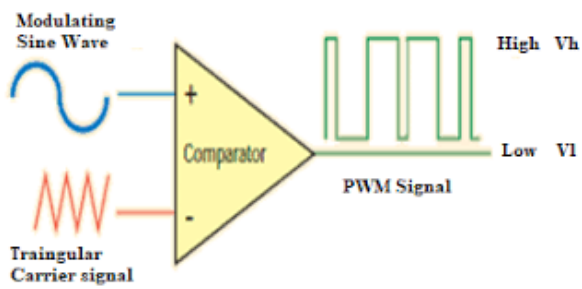

Figure 1.1: Generation of Pulse Width Modulation

1.2 Bipolar SPWM

Bipolar SPWM is used to generate sinusoidal output in single-phase full-bridge inverters. Its operation:

- Compare a sinusoidal reference (v_{sine}) with a triangular carrier (v_{tri}).
- Switch output between $+V_{\text{DC}}$ and $-V_{\text{DC}}$:

$$v_o = \begin{cases} +V_{\text{DC}} & \text{if } v_{\text{sine}} > v_{\text{tri}} \\ -V_{\text{DC}} & \text{if } v_{\text{sine}} < v_{\text{tri}} \end{cases} \tag{1.2}$$

- MOSFETs are controlled according to this logic to produce the desired AC waveform.

This method allows efficient voltage modulation and reduces harmonic distortion in the output, especially at higher modulation indices.'

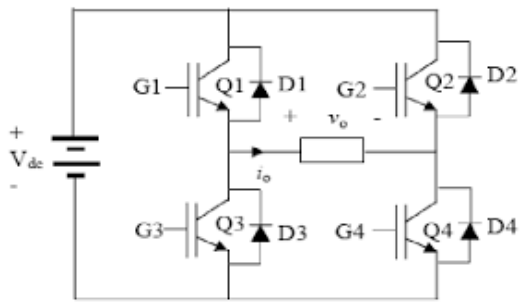

Figure 1.2: Full Bridge Circuit

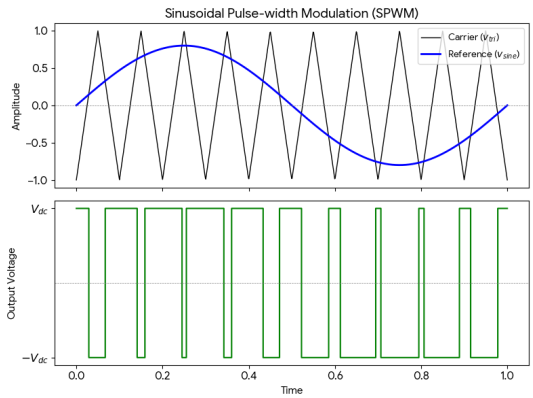

Figure 1.3: Bipolar PWM

1.3 Objective

The aim of this project is to:

- Simulate a single-phase H-Bridge inverter using SPWM in MATLAB Simulink.
- Analyze the effect of modulation index and PWM frequency on output voltage quality.
- Evaluate Total Harmonic Distortion (THD) to determine optimal operating conditions.

Chapter 2

Analysis and Simulation

2.1 Theoretical Analysis

In Sinusoidal Pulse Width Modulation (SPWM), a low-frequency reference sine wave of amplitude V_{ref} is compared against a high-frequency triangular carrier wave of amplitude V_{carrier}. The resulting PWM pulses drive the switches of an H-bridge inverter, synthesizing an AC output whose envelope follows the reference sine wave.

The key parameter is the modulation index M_a, defined as:

$$M_a = \frac{V_{\text{ref}}}{V_{\text{carrier}}} \tag{2.1}$$

In this simulation, $V_{\text{carrier}} = 1$ V, hence:

$$M_a = V_{\text{ref}} \tag{2.2}$$

The fundamental output peak voltage is given by:

$$V_{\text{out,peak}} = M_a \cdot V_{\text{DC}} \tag{2.3}$$

where $V_{\text{DC}} = 350$ V. For a desired RMS output voltage of 220 V:

$$V_{\text{out,peak}} = \sqrt{2} \times 220 \approx 311.1 \text{ V} \tag{2.4}$$

Thus, the required modulation index is:

$$M_a = \frac{311.1}{350} \approx 0.889 \tag{2.5}$$

2.2 MATLAB/Simulink Model Implementation

The SPWM inverter model was developed in MATLAB/Simulink. The implementation details and key waveforms are described below.

2.2.1 Circuit Configuration

The complete inverter circuit modeled in Simulink is shown in Figure 2.1. The system consists of a DC supply, an H-bridge with four MOSFET switches, and a resistive load.

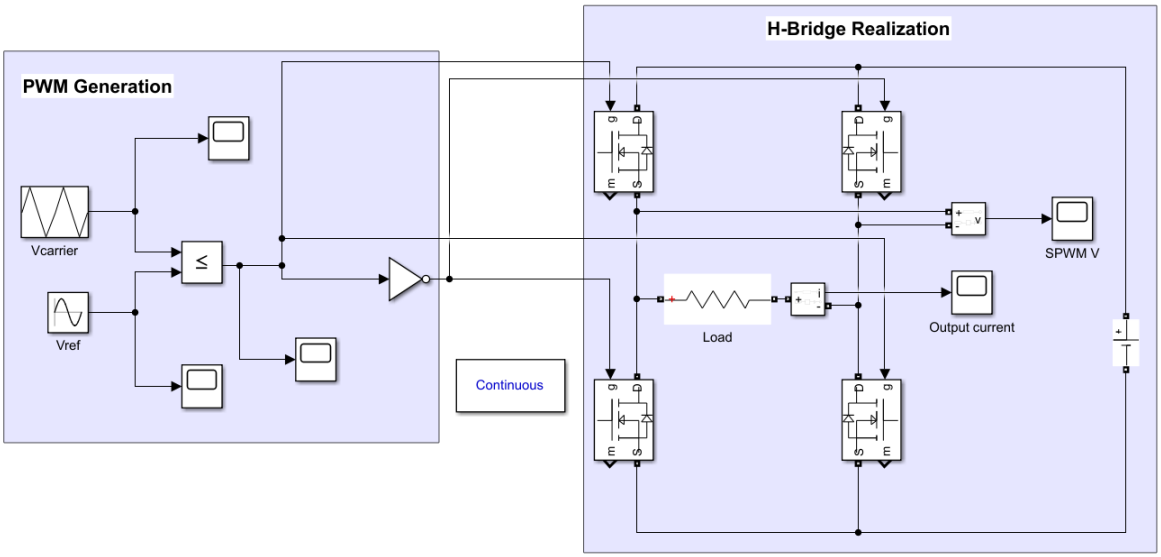

Figure 2.1: SPWM Inverter circuit modeled in Simulink

A constant DC input voltage of $V_{\mathrm{DC}} = 320\,\mathrm{V}$ is applied across the bridge, as shown in Figure 2.2, and a $100\,\Omega$ resistive load is connected at the output.

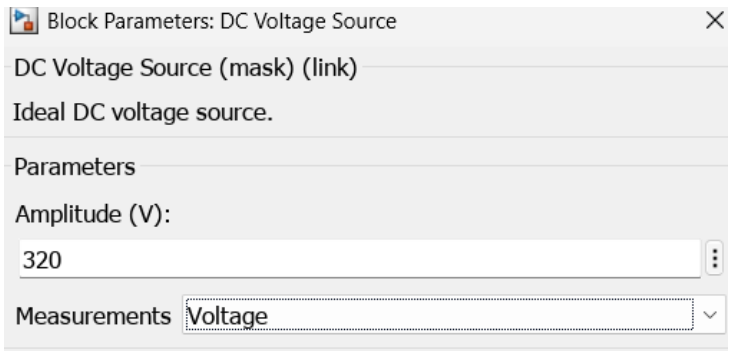

Figure 2.2: Input DC Voltage Supply

2.2.2 Signal Generation and PWM Formation

The SPWM technique is implemented by comparing a low-frequency sinusoidal reference signal with a high-frequency triangular carrier wave to generate the switching pulses.

- **Reference Signal:** A 50 Hz sine wave whose amplitude is varied from 0.1 to 1.5 to control the modulation index (Figure 2.4).
- **Carrier Signal:** A triangular wave with a fixed amplitude of 1 V, tested at frequencies of 5 kHz, 10 kHz, 15 kHz, and 20 kHz (Figure 2.3).

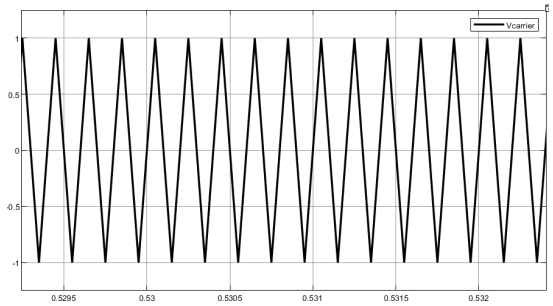

Figure 2.3: Carrier (Triangular) Signal

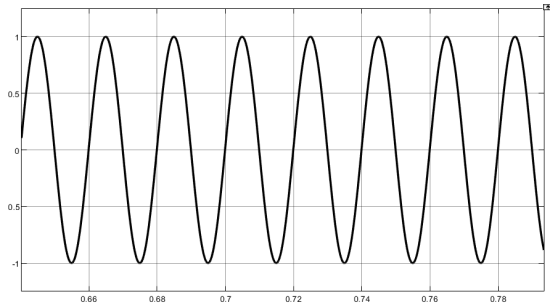

Figure 2.4: Reference (Sine Wave) Signal

2.2.3 Gating Signal Generation

The PWM pulses are generated by comparing the sine and triangular signals using a comparator block. The output of the comparator and its logical inverse (obtained via a NOT block) are used to drive the H-bridge switches.

- Original SPWM signal → Switches S_1 and S_4
- Inverted SPWM signal → Switches S_2 and S_3

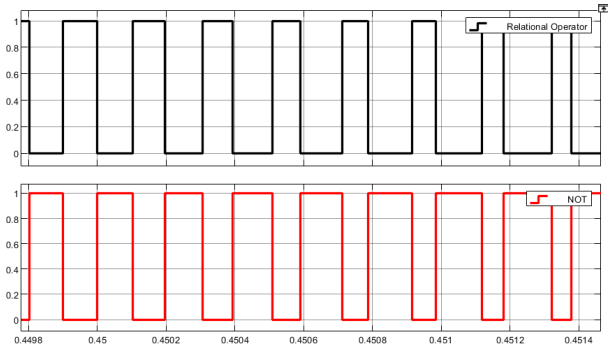

Figure 2.5: Comparator Output (PWM Signal)

Finally, the output voltage and current are measured using Scope blocks, and an FFT analysis is performed via the Powergui block to evaluate the Total Harmonic Distortion (THD).

Chapter 3

Results and Performance Analysis of SPWM Inverter

3.1 Load Voltage and Current Analysis

For a purely resistive load of $100\,\Omega$, the output current can be calculated using Ohm's law:

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{220}{100} = 2.2\,\text{A} \tag{3.1}$$

The peak current is given by:

$$I_{\text{peak}} = \sqrt{2} \times I_{\text{rms}} \approx 3.11\,\text{A} \tag{3.2}$$

Figure 3.1 shows the output voltage waveform of the SPWM inverter across the resistive load. The waveform consists of high-frequency PWM pulses whose average envelope follows a sinusoidal shape.

Figure 3.2 illustrates the corresponding load current. Since the load is purely resistive, the current waveform closely follows the voltage waveform, resulting in a sinusoidal shape with the same phase.

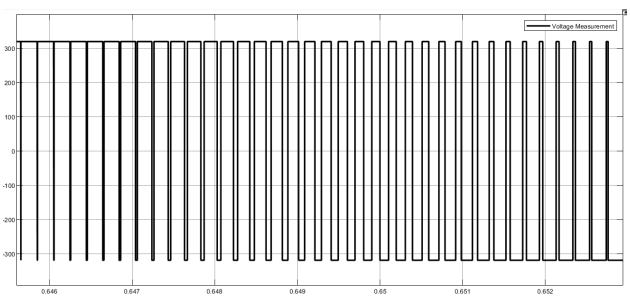

Figure 3.1: Output Voltage across Resistive Load

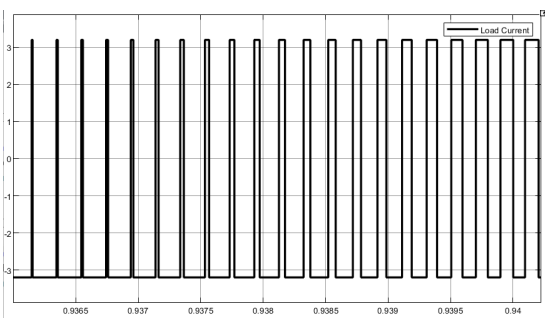

Figure 3.2: Load Current through the Resistive Load

3.2 Harmonic Analysis at Different Carrier Frequencies

The performance of the inverter was evaluated by varying the modulation index (M) from 0.1 to 1.5 across five different carrier frequencies. A summary of the combined results is presented in Figures 3.3 and 3.4, followed by the detailed numerical data in Table 3.1.

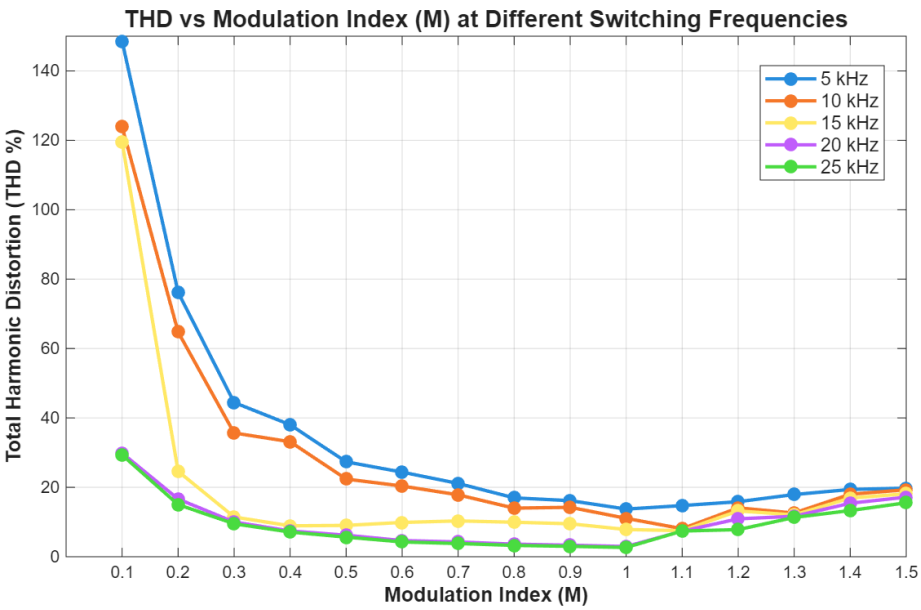

Figure 3.3: THD Analysis at Different Carrier Frequencies

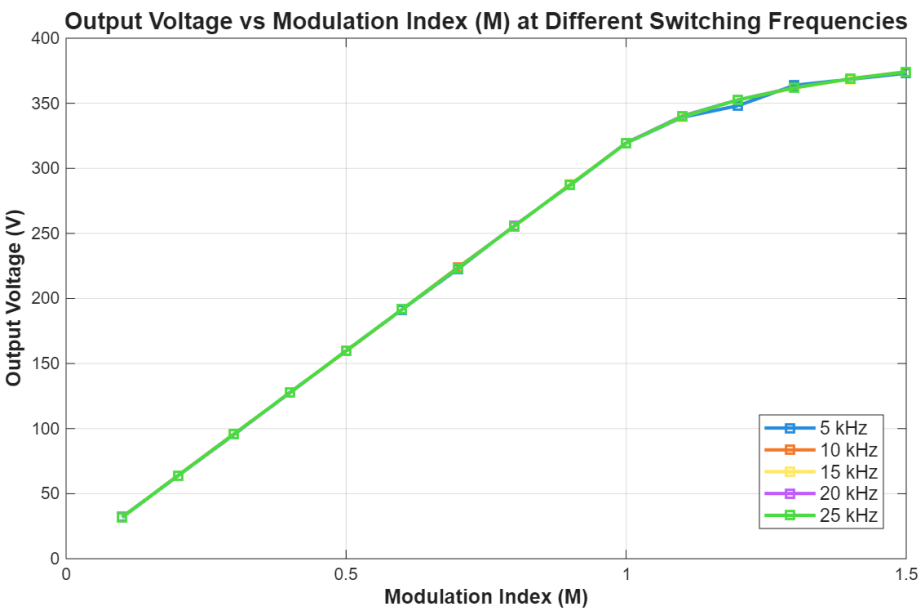

Figure 3.4: Fundamental Voltage Analysis at Different Carrier Frequencies

Table 3.1: THD and Output Voltage vs Modulation Index at Various Carrier Frequencies

	5kHz		**10kHz**		**15kHz**		**20kHz**		**25kHz**	
M	**THD**	**Voltage**	**THD**	**Voltage**	**THD**	**Voltage**	**THD**	**Voltage**	**THD**	**Voltage**
0.1	148.56	31.68	123.9	31.99	119.47	31.86	29.86	32.15	29.32	32.01
0.2	76.19	63.82	64.76	64.05	24.61	63.87	16.63	63.85	15.02	63.88
0.3	44.45	95.45	35.66	96.01	11.45	95.67	9.99	95.77	9.49	95.91
0.4	38.03	128	33.14	127.7	8.9	127.8	7.43	127.6	7.11	127.8
0.5	27.34	159.6	22.38	159.6	9.03	159.9	6.2	159.9	5.65	159.8
0.6	24.38	191.5	20.37	191.7	9.88	191.6	4.66	191.7	4.26	191.7
0.7	21.12	222.9	17.84	223.9	10.32	223.5	4.26	223.5	3.8	223.5
0.8	16.96	255.8	14	255.4	9.94	255.4	3.62	255.7	3.26	255.5
0.9	16.16	287.1	14.24	287.7	9.5	287.6	3.32	287.3	3	287.4
1.0	13.73	319.4	11.06	319.3	7.84	319.5	2.95	319.6	2.66	319.3
1.1	14.74	339.2	8.08	339.9	7.51	339.6	7.37	340.2	7.41	339.9
1.2	15.86	348.2	14.08	352.6	13.2	352.7	10.92	352.7	7.81	352.6
1.3	17.92	363.8	12.59	361.7	12.06	361.8	11.65	362.1	11.32	361.9
1.4	19.39	368.4	18.01	368.9	16.91	368.6	15.42	368.7	13.31	368.8
1.5	19.77	373.1	19.3	374.2	18.15	374.1	17.17	374	15.55	374.1

3.2.1 Comparison of THD and Fundamental Voltage

Table 3.2 summarizes the optimal operating points extracted from the data. It should be noted that modulation indices greater than 1.0 ($M > 1.0$) represent the over-modulation region, where the fundamental voltage increases slightly but at the cost of introducing low-order harmonics.

Table 3.2: Optimal THD and Corresponding Fundamental Voltage at Different Carrier Frequencies

Frequency	5kHz	10kHz	15kHz	20kHz	25kHz
M_a	1.0	1.1	1.1	1.0	1.0
THD (%)	13.73	8.08	7.51	2.95	2.66
Fundamental (V)	319.4	339.9	339.6	319.6	319.3

3.3 FFT Analysis

The FFT analyzer in Simulink is used to evaluate the spectral content of the inverter output voltage. It provides insight into the distribution of harmonic components and helps quantify THD under different operating conditions. The following figures present the FFT spectra corresponding to the optimal operating points identified in Table 3.2 for each carrier frequency. Figure 3.5 shows the spectrum for the best overall case, achieving a THD of 2.66% at a modulation index of 1.0 and a carrier frequency of 25kHz. Figures 3.6 through 3.9 illustrate the harmonic profiles at 5kHz, 10kHz, 15kHz, and 20kHz respectively.

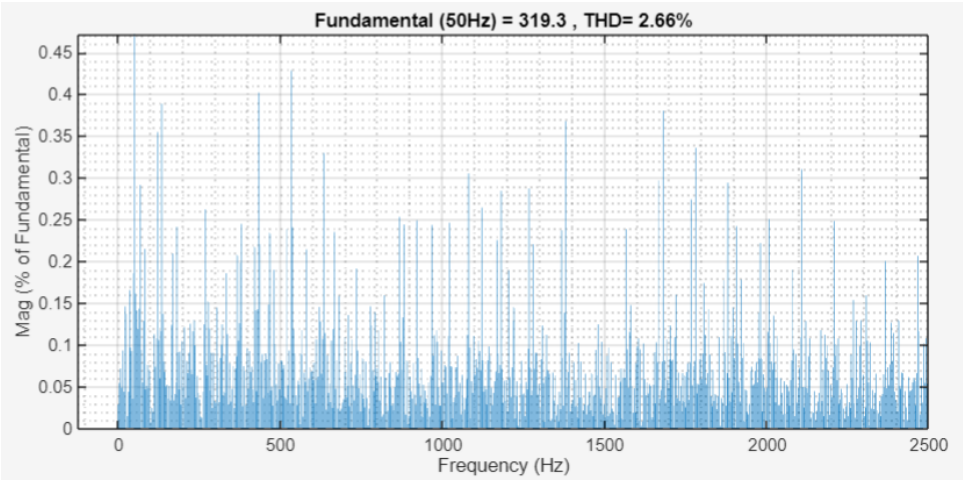

Figure 3.5: FFT Spectrum at 25kHz ($M = 1.0$, THD $= 2.66\%$)

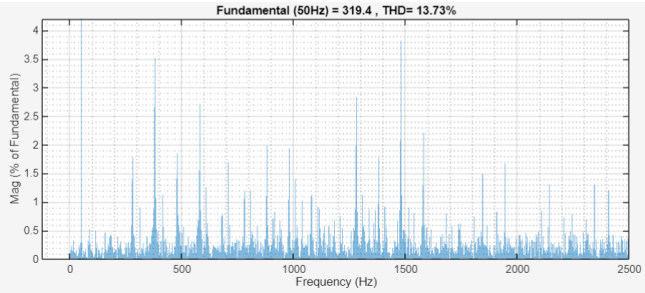

Figure 3.6: FFT Spectrum at 5kHz ($M = 1.0$, THD $= 13.73\%$)

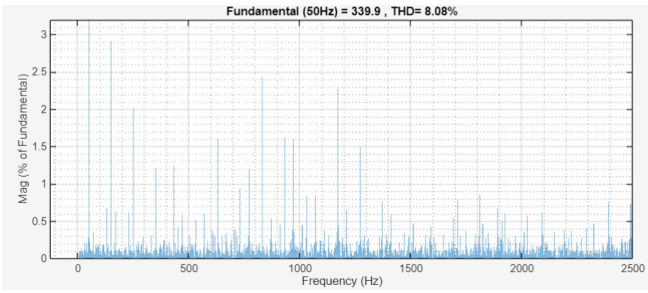

Figure 3.7: FFT Spectrum at 10kHz ($M = 1.1$, THD $= 8.08\%$)

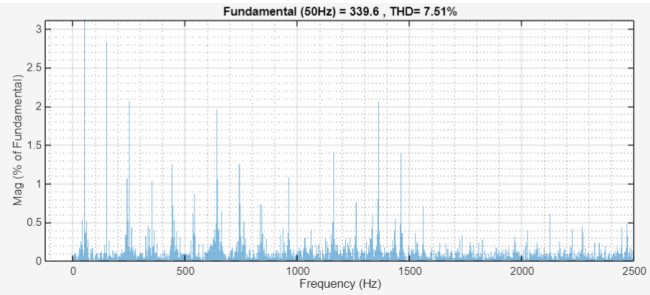

Figure 3.8: FFT Spectrum at 15kHz ($M = 1.1$, THD $= 7.51\%$)

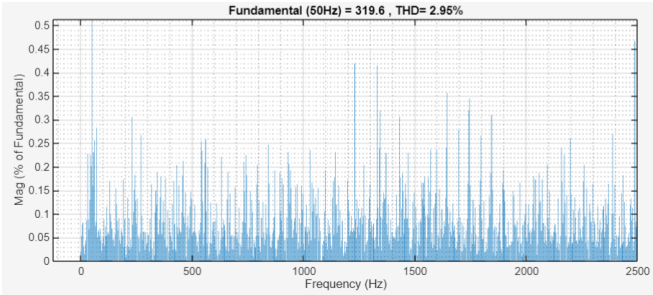

Figure 3.9: FFT Spectrum at 20kHz ($M = 1.0$, THD $= 2.95\%$)

Chapter 4

Discussion

4.1 Effect of M_a and Carrier Frequency on THD

The THD profile shows how output quality varies with M_a across different carrier frequencies. At low M_a, THD is high due to narrow pulse width of PWM. As M_a increases, the THD decreases steadily and reaches a minimum near $M_a = 1.0$. Beyond this point, the inverter enters the overmodulation region and THD begins to rise again due to waveform clipping.

Higher carrier frequencies consistently produce lower THD values across most of the modulation range. For instance, at $M_a = 1.0$, the THD drops from 13.73% at 5kHz to just 2.66% at 25kHz. This confirms that increasing the switching frequency pushes harmonic energy further away from the fundamental, resulting in a cleaner output.

However, higher switching frequencies increase switching losses in practical hardware. While 25kHz yields the lowest THD, 15–20kHz offers a practical compromise between harmonic performance and efficiency.

4.2 Optimal Configuration and Spectrum

As presented in the previous chapter, the best overall performance was achieved at $M_a = 1.0$ with 25kHz carrier, yielding THD = 2.66% and fundamental voltage $\approx$ 319 V. The corresponding FFT spectrum confirmed a dominant 50Hz fundamental with harmonic sidebands concentrated around multiples of the switching frequency. For practical applications where switching losses are a concern, a carrier frequency of 20kHz offers comparable results with a THD of 2.95%.

4.3 Overmodulation Considerations

Operating in the overmodulation region ($M_a > 1.0$) allows for a slight increase in fundamental output voltage without raising the DC bus voltage. However, this benefit comes at the cost of increased THD due to waveform clipping. From the data, a modulation index range of $1.0 < M_a \leq 1.15$ appears tolerable, as the THD remains below approximately 8% at higher carrier frequencies. Beyond $M_a = 1.2$, the distortion increases sharply, making the output unsuitable for applications requiring a clean sinusoidal waveform.

Chapter 5

Conclusion

This study examined the performance of a single-phase SPWM inverter by varying the modulation index (M_a) from 0.1 to 1.5 across carrier frequencies of 5 kHz, 10 kHz, 15 kHz, 20 kHz, and 25 kHz. The results confirm that both M_a and carrier frequency significantly influence the output waveform quality and harmonic content. The Total Harmonic Distortion (THD) decreases as M_a increases within the linear modulation region, reaching a minimum near $M_a = 1.0$. Beyond this point, overmodulation introduces waveform clipping and causes THD to rise, despite a marginal increase in fundamental voltage. Higher carrier frequencies consistently shift harmonic components away from the fundamental 50 Hz component, resulting in a cleaner output spectrum. The optimal performance was recorded at a modulation index of $M_a = 1.0$ combined with a carrier frequency of 25 kHz, yielding a THD of 2.66% and a fundamental voltage of approximately 319 V. For practical applications where switching losses must be considered, a carrier frequency of 20 kHz provides a suitable compromise with a THD of 2.95%.